A Survey on Hyperparameter Optimization of Machine Learning Models

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Abstract— **Hyperparameters in machine learning are those variables that are set before the training process starts and regulate several aspects of the behavior of the learning algorithm. In contrast to model parameters, which are determined by data during training, hyperparameters are outside factors that affect how the model discovers and generalizes patterns from the data. The selection of hyperparameters can have a considerable impact on a model's performance, rate of convergence, and ability to prevent overfitting. An effective hyperparameter optimization strategy can improve the performance of a machine learning model. A number of hyperparameter optimization techniques for different machine learning models are reviewed in this paper, including grid search, random search, Bayesian optimization, and genetic algorithm. Along with this different Evaluation tools and datasets used for machine learning model hyperparameter optimization in literature are examined. Various machine learning models like SVM, Naïve Baye's, KNN, ANN etc. whose hyperparameters are optimized in literature are also reported.**

Keywords—Hyperparameter tuning, Grid Search, random search, parameters, machine learning, optimization

I. INTRODUCTION

Machine learning, an innovative branch of artificial intelligence, has enabled computers to learn from data and improve their performance consistently. Machine learning gives computers the ability to spot patterns, make decisions that are well-informed, and derive insights from huge and complicated information. Whether it is used for natural language processing, computer vision, recommendation systems, or predictive analytics, machine learning has the power to reveal hidden relationships in data that could evade human study [1]. As a result, it has found use in a variety of fields, revolutionizing businesses and spurring innovation. Machine learning is at the cutting edge of technological development, changing how we engage with technology and opening up new possibilities through its ability to unearth important information and automate complex activities.

Model parameters and hyperparameters are the two primary categories of parameters that play different roles in the learning process in machine learning [2]. Model parameters are the values that are discovered by the machine learning algorithm from the training set of data. They specify the inner workings of the model and are iteratively modified using optimization algorithms to reduce the discrepancy between the model's forecasts and the actual target values in the training set. The model's capacity for precise prediction is directly influenced by these variables. For example, the weights and biases of the neurons in each layer make up the model parameters in neural networks [3]. On the other hand, hyperparameters are settings that are decided upon before to the start of the training process and regulate how the learning process itself is conducted. They are not determined by data learning, but rather by user selection or through a search procedure (hyperparameter tuning). Hyperparameters change the structure of the model and the behavior of the learning regularization, convergence speed, and generalization performance [4].

In machine learning, the act of selecting the appropriate set of hyperparameters for a particular model to achieve the best performance on a particular task or dataset is referred to as hyperparameter tuning [5]. Prior to the start of learning process, it is essential to provide hyperparameters values to control Examples of these parameters include the model's architecture, learning rate, number of hidden units in a neural network, regularization strength, and others [2]. In machine learning, hyperparameter optimization is a systematic procedure that entails a series of calculated actions intended to enhance model performance. Choosing a suitable machine learning model sets the groundwork for further hyperparameter exploration at the beginning of the trip [6].

Cross-validation is a crucial approach that gives the resilience of evaluation process and objectivity in the area of hyperparameter tweaking in machine learning. Cross-validation, which divides the dataset into multiple subsets or folds, ensures an impartial evaluation of a model's performance while the search for the best hyperparameters progress. Each fold serves as a validation set alternately during an iterative trip, while the others combine to create the training set. The model is trained and its performance is painstakingly assessed on the validation fold under the guidance of the chosen hyperparameters. A thorough set of performance indicators that accurately represent the behavior of the model across diverse data segments is the result of this iterative approach [7].

The optimal hyperparameter values for a specific dataset have previously been found through manual research. To set these settings, researchers usually reflect on their prior experience in training these machine learning algorithms to handle issues similar to those discussed [8]. Unfortunately, the best hyperparameter settings for problems might not be the best settings for another because these values differ across datasets. Consequently, it is challenging to specify the hyperparameter values based on prior knowledge. The classification accuracy of the algorithm typically determines how well a hyperparameter performs.

An essential component of the algorithm's performance is its precision. However, one performance metric to take into account while applying is the verification time [9]. Datasets generally used for hyperparameter optimization are MNIST, Fashion MNIST having configuration 6000,60000 grey scale

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images(28x 28), 600000 RGB images(32x32), 40 to 800 images(300x200)[9], CIFAR-10 having configuration 60,000 color pictures(32x32x3), 70,000(48x48x3), CIFAR-100[10].

In this paper we have reviewed various hyperparameter optimization techniques for machine learning models like grid search, random search, Bayesian optimization, genetic algorithm on the basis of their characteristics, their advantages and disadvantages and key findings. These hyperparameter optimization methods address the difficulties brought on by the high-dimensional, complicated spaces of hyperparameters by striking a compromise between thoroughness and effectiveness. Grid Search encompasses every possibility, but it could be computationally taxing [10]. Random Search strikes a mix between efficiency and exploration, Bayesian Optimization is effective for expensive objective functions, and Genetic Algorithms are appropriate for difficult search spaces [11]. Study of Hyperparameter optimizations in machine learning is the focus of the work that is being presented. The goal of this study is to give a comprehensive analysis of the current approaches being used in this field. With a focus on the datasets and assessment criteria frequently used in these optimization procedures, it attempts to explore the optimization of hyperparameters across a range of machine learning models. The paper also explores the applicability of hyperparameter optimization strategies across several model contexts in order to highlight the flexibility and adaptability of these techniques.

This work is divided into a number of sections in section 2, we will give general review of the methods and tactics employed in current approaches. A detailed analysis of hyperparameter optimization methods for various machine learning models is the focus of Section 3. In section 4, attention is given to the datasets and assessment tools used in the context of hyperparameter optimization. This section examines the effects of data attributes and evaluation standards on the optimization process. The paper's concluding portion is discussed in section 5 makes judgments based on the knowledge gained from the earlier sections. In addition, this section identifies prospective directions for additional study and advancement in the area.

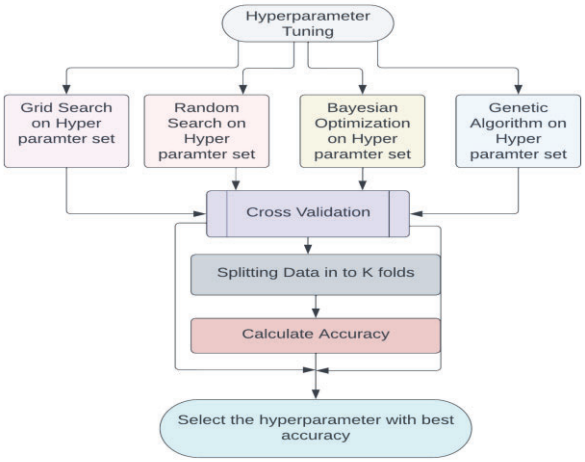


Fig. 1. Techniques of hyper parameter tuning

II. Existing Techniques of Machine Learning Hyperparameter Optimization

Optimizing the configuration of hyperparameters to achieve the greatest model performance is known as "hyperparameter optimization," and it is a critical activity in the field of machine learning. These techniques' complexity is geared toward navigating the complex environment of hyperparameter spaces, where each set of variables affects a model's learning dynamics and eventual predictive power [6]. Grid search, random search, Bayesian optimization, genetic algorithm, and particle swarm optimization are some of the approaches. Numerous studies are conducted which provides that grid search explores preset hyperparameter ranges systematically, whereas random search stochastically examines configurations of considered model. Figure 1 shows how hyperparameters are tuned using optimization techniques like grid search, random search, Bayesian optimization, genetic algorithm, then cross validation is applied, which requires dividing the dataset in to numerous folds known as subsets. Accuracy is calculated by applying algorithms, on the basis of best accuracy hyperparameters are selected and this process repeats itself.

The Bayesian theorem, which is the foundation of Bayesian optimization, creates a prior over the optimization function and updates the posterior of the optimization function using the information from the previous collection of data [13]. Bayesian optimization skillfully directs the search into promising areas by utilizing statistical modeling to wisely balance exploration and exploitation. Biological evolution served as the inspiration for evolutionary algorithms, which simulate populations of hyperparameter sets to converge on optimal values through selective adaptation [14].

TABLE I. COMPRISION OF HYPEROPTIMIZATION TECHNIQUE

S no	**Technique**	**Characteristics**	**Advantages**	**Disadvantages**	**Key Findings**
1	Grid Search[15]	Searching exhaustively through a hyperparameter grid that has been set up	Easy to use and comprehend	For huge search spaces, computationally expensive	Appropriate for compact search areas
		Reviews all possible combinations of hyperparameters	Guarantees locating the ideal response	Effectiveness for continuous hyperparameters is poor	Poor performance in high-dimensional spaces is a tendency
		For distributed evaluation, parallelization is straightforward	Replicable outcomes without chance		Ideal for tiny hyperparameter search spaces
2	Random Search[15]	Using a given space as a sample, randomly selecting the		There is no assurance that the best option will be	not all possible combinations of hyperparameters are evaluated.

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		hyperparameters		discovered.	
		a distributed evaluation that is simple to parallelize	explores a variety of areas in the search space.		Shows better performance compared to Grid Search when computational resources are limited
3	Genetic Algorithm[9]	Population -based approach, variation through cross over & Mutation	Global search capability, huge search space exploration, overfitting is avoided	Computationally Expensive, a scant investigation of interactions' effects	Suitable for discrete hyper parameter, find intricate interactions between hyperparameters
4	Bayesian Optimization[1]	explores a variety of areas in the search space.	Effective for pricey and noisy tasks	higher computational requirements	converges faster and finds better solutions than Grid Search and Random Search.
		compromises between exploration and exploitation	faster discovery of ideal hyperparameters	adjustment of the acquisition function and surrogate model is necessary	It is the recommended solution for pricey optimization initiatives because of its capacity to examine the search field effectively.

Furthermore, automated machine learning (Auto ML) platforms provide a comprehensive strategy that includes model selection, feature engineering, and hyperparameter tuning [7]. These techniques, which are typically coupled and adjusted to the specific problem environment, are an excellent illustration of the complexity of hyperparameter tuning. In order to maximize the potential of machine learning models, an optimized values of hidden layers should be used in the configuration [8].

III. Review of Hyperparameter Optimization of different machine Learning models

There are several machine learning models which are all modeled to address variety of problems in different domains. Some of them are Linear Regression, Logistic regression, Random Forest, gradient boosting, SVM, Naïve Bayes, KNN etc. Data from the repository demonstrates that random forest is the best classifier for evaluating the performance [13]. Bayesian Optimization can be used for hyperparameter tunning of ANN & CNN. Grid search and genetic algorithm provide same accuracy using SVM and KNN. Genetic Algorithm uses Selection, crossover and mutation as evolutionary operators [1].As depicted in the figure 3,CNN has various parameters which need to tuned using classification and evolutionary algorithms.These are optimized using optimizers.

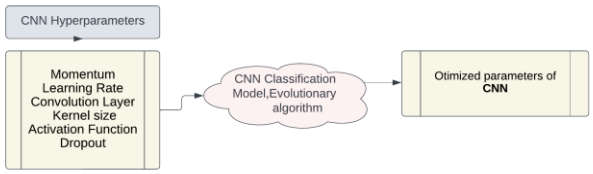


Fig. 2. CNN Hyperparameters

Different hyperparameter optimization algorithms are used to obtain suitable values of hyperparameters which are assessed through accuracy, momentum, learning rate, dropout etc. [10]. Hussain Alibrahim,Simone A. Ludwig (2021)proved that on applying grid search on SVM, Random forest, KNN, Random forest gives better results in terms of accuracy and time[10]. Researchers in 2012 proposed the idea of randomized search and illustrated its superiority over grid search in terms of accuracy [18]. Paper presented Bayesian optimization for hyperparameter tuning and showed how it might improve complex functions with a small number of evaluations [19].

Elgeldawi E, Sayed A, Zaki (2021) proposed a method to combine numerous optimization algorithms, showing the potential advantages of harnessing the collective strength of various techniques. Ensemble methods have also made their way into hyperparameter optimization [20]. Grid search, random search, Bayesian optimization, evolutionary algorithms, and other techniques were all thoroughly examined and designated as hyperparameter optimization methods in the survey [21]. In comparison to related techniques like Multi-Task Bayesian optimization, Hyperband, and ordinary Bayesian optimization, FABOLAS typically generated effective designs 10 to 100 times faster [22]. Jia Wu, Xiu-Yun Chen, Hao Zhang (2019) showed that by using deep neural networks and support vector machines, the Support Vector classifier obtained the greatest Bayesian Optimization score (95.6208), having the best accuracy both before and after the hyperparameter optimization. Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), meanwhile, significantly raised the Naive Bayes classifier's score [23]. Baosu Guo, Jingwen Hu(2019) shown that by using a grid search strategy, the recommended method improved the hyperparameters for the prediction algorithms. Accuracy, precision, recall, the f1-score, the AUC-ROC, the MAE, the RMSE, the R2, and the confusion matrix are a few commonly used metrics that are used to compare the outcomes of each trial and evaluate the effectiveness of the method. Application of the suggested 10-fold cross validation and GSHPO procedures yielded encouraging results [2].Key findings imply that tweaking model hyper-parameters improve model prediction accuracy in a statistically significant way.

According to the results published in the paper [10], genetic algorithm is applied for hyperparameter optimization of various machine learning modes. Out of different models compared, random forest proved to be better in terms of

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accuracy. A series of calculated actions targeted at improving a model's performance comprise the machine learning process of hyperparameter optimization. First, the potential ranges or values for each hyperparameter are determined, defining the operating range for them [24]. These hyperparameters include crucial settings like learning rates, regularization strengths, hidden layer counts, and others that have a direct impact on the model's learning behavior. The next stage is to choose an exploration plan after this space has been identified. The evaluation of various hyperparameter combinations to determine the best configuration is governed by this technique. Commonly utilized techniques include random search, which samples configurations at random, grid search, which evaluates every conceivable combination exhaustively, and Bayesian optimization, which intelligently directs the search process [12].

To improve model performance, the hyperparameter tuning procedure entails a series of well-planned stages. In the pursuit of more accurate machine learning models, methods like random search, grid search, and Bayesian optimization have all become useful tools for navigating the hyperparameter field.

A crucial component of optimizing machine learning models for top performance is hyperparameter tuning. It entails a methodical procedure that starts with choosing hyperparameters pertinent to the particular algorithm under consideration. The crucial next step is to define a search space, which is frequently indicated by allowable ranges or values for each hyperparameter. In order to efficiently navigate the search space, one can then select an optimization strategy, such as grid search, random search, Bayesian optimizations, or genetic algorithms. The performance of the model is then evaluated using the metrics established. The validation set serves as the testing ground for different hyperparameter settings, and the data is divided into training, validation, and test sets. Hyperparameter optimization seeks to identify the optimal configuration through iterative modifications. The model is then tested on an independent test set before deployment to ensure robust generalization, and the best configuration is chosen based on the findings from the validation set. The procedure is shown in figure 2.

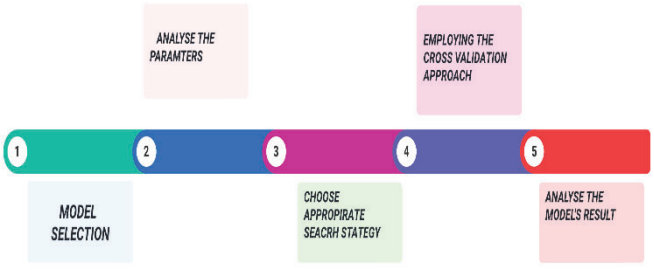


Fig. 3. Steps to perform Hyperparameter tuning

IV. DATASET AND EVALUATION MEASURES USED ACROSS HYPERPARAMETER OPTIMIZATION OF MACHINE LEARNING MODELS

Various Studies are analyzed to understand the universality among all algorithms. Gird Search, Random Search, and Bayesian optimization techniques have all successfully tackled the hyper-parameter optimization problem of learning algorithms. They are regularly used as organic benchmarks. The hyper-parameter optimization problem of learning algorithms is also addressed by the optimization strategy based on heuristic algorithms and other ways recently developed, although their performance is more varied and most of them are only effective in a limited number of circumstances. Researchers have evaluated the performance of hyperparameter optimization algorithms on majorly datasets like MNIST,Fashion MNIST Dataset,CIFAR-10,CIFAR-100[22],[23]. The optimization done through genetic algorithm, Grid search, Random search, Bayesian optimization are evaluated on the basis of time taken by these algorithms to calculate the accuracy. The decision is based on the nature of the problem, the computational resources at hand, and the required level of optimization efficiency. Grid search and random search can still be useful for simpler issues or when processing resources are limited, but Bayesian optimization and genetic algorithms tend to be more effective and adaptable for bigger and more complex search areas [5].

Hyperparameter trade-offs occur frequently. For instance, boosting the learning rate could hasten convergence but also raise the possibility of exceeding the ideal answer. Heuristic algorithms and innovative approaches have been used to tackle the challenge of hyperparameter optimization; however, these techniques frequently have inconsistent performance across a wide range and are limited to a few problem domains. On key datasets including MNIST, Fashion MNIST, CIFAR-10, and CIFAR-100, researchers have thoroughly assessed the effectiveness of hyperparameter optimization strategies

V. CONCLUSION AND FUTURE SCOPE

Hyperparameter are one of the factors on which the efficacy of any machine learning model depends. Hence its optimization is an important task before applying any machine model on application. Applying a potent hyperparameter optimization strategy can increase the efficacy of a machine learning model. In the work presented here, a survey of machine learning model hyperparameter optimization has been conducted. Numerous hyperparameter optimization techniques, such as grid search, random search, Bayesian optimization, and evolutionary algorithm, have all been reviewed. Additionally, several assessment metrics and datasets employed. The literature is examined for hyperparameter optimization of machine learning models. We also report on a number of machine learning models, including SVM, Naive Baye's, KNN, ANN, etc., whose hyperparameters have been optimized in the literature.

In the future, multi-objective particle swarm optimization, non-dominated sorting genetic algorithm (NSGA-2), and a number of other meta-heuristic techniques can be used to optimize the hyperparameters of machine learning models. Quantum-based meta-heuristic algorithms can be utilized to get better values for the hyperparameters of machine learning and deep learning models.

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